Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Which of the following arrangements represents the increasing order (smallest to largest) of ionic radii of the given species $O^{2-}, S^{2-}, N^{3-}, P^{3-}$?

(A) $O^{2-} < N^{3-} < S^{2-} < P^{3-}$

(B) $O^{2-} < P^{3-} < N^{3-} < S^{2-}$

(C) $N^{3} < O^{2-} < P^{3-} < S^{2-}$

(D) $N^{3-} < S^{2-} < O^{2-} < P^{3-}$

Q.2

The formation of the oxide ion $O^{2-}(g)$, from oxygen atom requires first an exothermic and then an endothermic step as shown below:
$O(g) + e^{-} \rightarrow O^{-}(g); \Delta_f H^{\ominus} = -141 \text{ kJ mol}^{-1}$
$O^{-}(g) + e^{-} \rightarrow O^{2-}(g); \Delta_f H^{\ominus} = +780 \text{ kJ mol}^{-1}$
Thus process of formation of O^{2-} in gas phase is unfavourable even though O^{2-} is isoelectronic with neon. It is due to the fact that

(A) Electron repulsion outweighs the stability gained by achieving noble gas configuration

(B) O^{-} ion has comparatively smaller size than oxygen atom

(C) Oxygen is more electronegative

(D) Addition of electron in oxygen results in larger size of the ion.

Q.3

Consider the following statements:
(i) Atomic radii decreases across a row of the periodic table when we move from left to right.
(ii) Atomic radii increases down the column as we move from top to bottom.
(iii) Although the order of elements is based on atomic numbers, vertical families share similar chemical properties. Which of the statement(s) given above is/are correct?

(A) (i) and (ii)

(B) (i) and (iii)

(C) (ii) and (iii)

(D) (i), (ii) and (iii)

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Q.4

Consider the following changes

$A \rightarrow A^{+} + e^{-} : E_1$ and $A^{+} \rightarrow A^{2+} + e^{-} : E_2$

The energy required to pull out the two electrons are E_1 and E_2 respectively. The correct relationship between two energies would be

(A) $E_1 < E_2$

(B) $E_1 = E_2$

(C) $E_1 > E_2$

(D) $E_1 \geq E_2$

Q.5

Ionization energies of atoms A and B are 400 and $300k$ cal mol^{-1} respectively. The electron affinities of these atoms are 80.0 and $85.0k$ cal mol^{-1} respectively. Then which is the correct statement regarding electronegativity χ?

(A) $\chi_A < \chi_B$

(B) $\chi_A > \chi_B$

(C) $\chi_A = \chi_B$

(D) none of these

Q.6

In which of the following process highest energy is absorbed?

(A) $Cu \rightarrow Cu^{+}$

(B) $Br \rightarrow Br^{-}$

(C) $I \rightarrow I^{-}$

(D) $Li \rightarrow Li^{+}$

Q.7

An element of atomic weight 40 has $2, 8, 8, 2$ as the electronic configuration. Which one of the following statements regarding this element is not correct?

(A) it belongs to II group of the periodic table

(B) it has 20 neutrons

(C) the formula of its oxide is MO_2

(D) it belongs to 4 th period of the periodic table

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Q.8

Which of the following series correctly represents relations between the elements from X to Y? $\mathrm{X} \rightarrow \mathrm{Y}$

(A) ${}_{3}\mathrm{Li} \rightarrow {}_{19}\mathrm{K}$ Ionization enthalpy increases

(B) ${}_{9}\mathrm{F} \rightarrow {}_{35}\mathrm{Br}$ Electron gain enthalpy (negative sign) increases

(C) ${}_{6}\mathrm{C} \rightarrow {}_{32}\mathrm{Ge}$ Atomic radii increases

(D) $18\mathrm{Ar} \rightarrow {}_{54}\mathrm{Xe}$ Noble character increases

Q.9

Which of the following arrangements represents the increasing order (smallest to largest) of ionic radii of the given species $\mathrm{O}^{2-}, \mathrm{S}^{2-}, \mathrm{N}^{3-}, \mathrm{P}^{3-}$?

(A) $\mathrm{O}^{2-} < \mathrm{N}^{3-} < \mathrm{S}^{2-} < \mathrm{P}^{3-}$

(B) $\mathrm{O}^{2-} < \mathrm{P}^{3-} < \mathrm{N}^{3-} < \mathrm{S}^{2-}$

(C) $\mathrm{N}^{3} < \mathrm{O}^{2-} < \mathrm{P}^{3-} < \mathrm{S}^{2-}$

(D) $\mathrm{N}^{3-} < \mathrm{S}^{2-} < \mathrm{O}^{2-} < \mathrm{P}^{3-}$

Q.10

Elements X, Y and Z have atomic numbers 19,37 and 55 respectively. Which of the following statements is true about them ?

(A) Their ionization potential would increase with increasing atomic number

(B) 'Y' would have an ionisation potential between those of 'X' and 'Z'

(C) 'Z' would have the highest ionization potential

(D) 'Y' would have the highest ionization potential.

Q.11

In which of the following arrangements, the sequence is not strictly according to the property written against it?

(A) $\mathrm{CO}_2 < \mathrm{SiO}_2 < \mathrm{SnO}_2 < \mathrm{PbO}_2$: increasing oxidising power

(B) $\mathrm{NH}_3 < \mathrm{PH}_3 < \mathrm{AsH}_3 < \mathrm{SbH}_3$: increasing basic strength

(C) $\mathrm{HF} < \mathrm{HCl} < \mathrm{HBr} < \mathrm{HI}$: increasing acid strength

(D) $\mathrm{B} < \mathrm{C} < \mathrm{O} < \mathrm{N}$: increasing first ionisation enthalpy.

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Q.12

The increasing order of electron affinity of N, P and As is

(A) $\mathrm{N} < \mathrm{P} < \mathrm{As}$

(B) $As < P < N$

(C) $As < N < P$

(D) $As < N > P$

Q.13

Which of the following order is wrong?

(A) $\mathrm{NH_3} < \mathrm{PH_3} < \mathrm{AsH_3}$ – Acidic

(B) $\mathrm{Li} < \mathrm{Be} < \mathrm{B} < \mathrm{C} - \mathrm{IE_1}$

(C) $\mathrm{Al_2O_3} < \mathrm{MgO} < \mathrm{Na_2O} < \mathrm{K_2O}$ – Basic

(D) $\mathrm{Li^+} < \mathrm{Na^+} < \mathrm{K^+} < \mathrm{Cs^+}$ – Ionic radius

Q.14

Which property decreases from left to right across the periodic table and increases from top to bottom?
(i) Atomic radius
(ii) Electronegativity
(iii) Ionisation energy
(iv) Metallic character

(A) (i) only

(B) (i), (ii) and (iii)

(C) (i), (iii) and (iv)

(D) (i) and (iv)

Q.15

Which alkali metal is most metallic in character

(A) K

(B) Cs

(C) Na

(D) Li

Q.16

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The atomic size of Ga is smaller than Al because

(A) Poor shielding effect of electrons in d-orbitals, effective nuclear charge increases in Gallium

(B) More shielding effect of d-electrons

(C) Poor shielding effect of f-orbitals in Gallium

(D) High shielding effect of $3\ \mathrm{s}, 3\mathrm{p}$ and $3\ \mathrm{d}$ electrons

Q.17

Which one of the following is an amphoteric oxide?

(A) MgO

(B) $\mathrm{Al_2O_3}$

(C) $\mathrm{NO_2}$

(D) $\mathrm{P_2O_5}$

Q.18

The formation of the oxide ion $\mathrm{O^{2-}(g)}$ requires first an exothermic and then an endothermic step as shown below. $\mathrm{O(g) + e^- = O^-(g);} \quad \Delta H^\circ = -142\ \mathrm{kJ\ mol^{-1}}$
$\mathrm{O^-(g) + e^- = O^{2-}(g);} \quad \Delta H^\circ = 844\ \mathrm{kJ\ mol^{-1}}$
This is because

(A) oxygen is more electronegative

(B) oxygen has high electron affinity

(C) $\mathrm{O^-}$ ion will tend to resist the addition of another electron

(D) $\mathrm{O^-}$ ion has comparatively larger size than oxygen atom.

Q.19

Which of the following series correctly represents relations between the elements from X to Y ? $\mathrm{X \rightarrow Y}$

(A) ${}_3\mathrm{Li} \rightarrow {}_{19}\ \mathrm{K}$ Ionization enthalpy increases

(B) ${}_9\ \mathrm{F} \rightarrow {}_{35}\mathrm{Br}$ Electron gain enthalpy with negative sign increases

(C) ${}_6\mathrm{C} \rightarrow {}_{32}\mathrm{Ge}$ Atomic radii increases

(D) ${}_{18}\mathrm{Ar} \rightarrow {}_{54}\mathrm{Xe}$ Nobel character increases

Q.20

Beryllium bears diagonal relationship with

(A) magnesium

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(B) boron

(C) aluminium

(D) lithium

Answers & Solutions

Q.1 Answer:

$O^{2-} < N^{3-} < S^{2-} < P^{3-}$

Solution:

For isoelectronic species ionic radii decreases as the charge on ion decreases. Further on moving down in a group ionic radii increases. Hence the correct order is
$O^{2-} < N^{3-} < S^{2-} < P^{3-}$

Q.2 Answer:

Electron repulsion outweighs the stability gained by achieving noble gas configuration

Solution:

Incoming electrons occupies the smaller $n = 2$ shell, also negative charge on oxygen (O^-) is another factor due to which incoming electron feel repulsion. Hence electron repulsion outweigh the stability gained by achieving noble gas configuration.

Q.3 Answer:

(i), (ii) and (iii)

Solution:

No solution available

Q.4 Answer:

$E_1 < E_2$

Solution:

IE_1 is always less than IE_2.

Q.5 Answer:

$\chi_A > \chi_B$

Solution:

Electronegativity of an atom $= \frac{IE+EA}{2\times 62.5}$
where IE and EA are in k cal mol $^{-1}$.

Q.6 Answer:

$Cu \rightarrow Cu^+$

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Solution:

Cu has completely filled d -orbital, so highest energy is absorbed when it converts to Cu^{+} ion.

Q.7 Answer:

the formula of its oxide is MO_2

Solution:

Its valency is 2 . So it will form MO type compound.

Q.8 Answer:

${}_6C \rightarrow {}_{32}$ Ge Atomic radii increases

Solution:

On moving down a group atomic radii increases.

Q.9 Answer:

$O^{2-} < N^{3-} < S^{2-} < P^{3-}$

Solution:

For isoelectronic species ionic radin decreases as the charge on ion decreases. Further on moving down in a group ionic radii increases. Hence the correct order is $O^{2-} < N^{3-} < S^{2-} < P^{3-}$

Q.10 Answer:

'Y' would have an ionisation potential between those of ' X ' and ' Z '

Solution:

Elements X, Y, Z with atomic numbers 19 , 37,55 lie in group 1 (alkali metals). On moving down a group from the size of atoms increases, the outermost electrons become less strongly held. So the ionization energy decreases. Therefore, IE of Y could be between those of X and Z.

X	Y	Z
$K(19)$	$Rb(37)$	$Cs(55)$
4.3	4.2	3.9

I.E 43

Q.11 Answer:

$NH_3 < PH_3 < AsH_3 < SbH_3$: increasing basic strength

Solution:

Correct order of increasing basic strength is $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$

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Q.12 Answer:

$As < N < P$

Solution:

Phosphorus has vacant ' d '-orbitals due to which it has higher electron affinity than nitrogen. As also has vacant d -orbitals but its atomic size is very big thus its electron affinity will be lesser than N .

Q.13 Answer:

$\mathrm{Li} < \mathrm{Be} < \mathrm{B} < \mathrm{C} - \mathrm{IE}_1$

Solution:

The right sequence of I.E $_1$ of $\mathrm{Li} < \mathrm{B} < \mathrm{Be} < \mathrm{C}$.

Q.14 Answer:

(i) and (iv)

Solution:

Atomic radius and metallic character decreases from left to right across the period and increases from top to bottom down the group.

Q.15 Answer:

Cs

Solution:

$\mathrm{Cs} > \mathrm{Rb} > \mathrm{K} > \mathrm{Na} > \mathrm{Li}$

Metallic character decreasing order

Q.16 Answer:

Poor shielding effect of electrons in d-orbitals, effective nuclear charge increases in Gallium

Solution:

Lesser the shielding effect, more will be force of attraction between valence electron and Incleus, smaller will be atomic size

Q.17 Answer:

$\mathrm{Al_2O_3}$

Solution:

$\mathrm{Al_2O_3}$ is an amphoteric oxide. It can react with an acid as well as with the base.

$\mathrm{Al_2O_3 + HCl \longrightarrow ACl_3 + H_2O Al_2O_3 + NaOH \longrightarrow NaAlO_2 + H_2O}$

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Q.18 Answer:

O^- ion will tend to resist the addition of another electron

Solution:

The addition of negatively-charged electron to the negatively-charged species O^- requires an input of energy.

Q.19 Answer:

${}_6C \rightarrow {}_{32}Ge$ Atomic radii increases

Solution:

On descending a group of the periodic table, atomic radii increases.

Q.20 Answer:

aluminium

Solution:

aluminium

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